

- * square each number
- * Add the squared deviation
- * divide the number of values

$$1-3 \rightarrow -2 \rightarrow (-2)^2 = 4$$

$$(2-3) \rightarrow (-1) \rightarrow (-1)^2 = 1$$

$$3-3 \rightarrow 0 \rightarrow 0$$

$$4-3 \rightarrow 1 \rightarrow 1$$

$$5-3 \rightarrow 2 \rightarrow 2^2 = 4$$

$$= 10 \quad 10/5 = 2$$

c) standard deviation:

It measures how much data value is different from others.

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

small static

$$\text{arr} = [1, 2, 3, 4, 5]$$

arr print ("static (static - sdev(arr))")

Unit - 3 Solved Problems.

① Descriptive Statistics

→ data summarization

→ mean, mode, median & Range

P.1: marks of 10 students are

45, 52, 60, 48, 75, 60, 55, 68, 72, 50

$$\text{mean} = \frac{\sum x}{n} = \frac{45 + 52 + 60 + 48 + 75 + 60 + 55 + 68 + 72 + 50}{10}$$

$$= 385/10 = 38.5$$

Step 2

$$\text{median} = 45, 48, 50, 52, 55, 60, 68, 72, 75$$

$$= 57.5$$

Step 3: mode = 60

$$\text{range} = 75 - 45 = 30$$

② the following are the number of hours (is standard, same) 2, 4, 3, 5, 2, 6, 4, 2, 5, 7, 2, 4, 6, 5, 3

sol given

2, 4, 3, 5, 2, 6, 4, 3, 5, 7, 2, 4, 6, 5, 3

step 1: Arrange the values
2, 3, 4, 5, 6, 7

step 2: count the frequency

Study hours	Frequency
2	3
3	2
4	3
5	3
6	2
7	1
Total	15

step 3: calculate percentage

$$\text{percentage} = \frac{F}{N} \times 100$$

$\therefore$ Frequencies

N = total number of observations = 15

Hours	Frequency	percentage
2	3	$3/15 \times 100 = 20\%$
3	2	$2/15 \times 100 = 13.33\%$
4	3	$3/15 \times 100 = 20\%$
5	3	$3/15 \times 100 = 20\%$
6	2	$2/15 \times 100 = 13.33\%$
7	1	$1/15 \times 100 = 6.67\%$
Total	15	100%

Answer: The highest frequency is 3 occurring for 2, 4, and 5 hours

③ consider the following dataset 10, 12, 15, 18, 20, 22, 28, 30, 35.

Find using IQR

sol/ given

10, 12, 15, 18, 20, 22, 28, 30, 35

$$n = 10$$

Find Q_1 : $n = 10$ the median is the avg of the 5th & 6th value

$$P_2 = \frac{20 + 22}{2} \quad Q_2 = 21$$

Find Q_1

10, 12, 15, 18, 20

$$Q_1 = 15$$

middle value

Find Q_3

upper half

22, 25, 28, 30, 35

the middle value

$$Q_3 = 28$$

calculate IQR

$$IQR = Q_3 - Q_1$$

$$IQR = 28 - 15$$

$$IQR = 13$$

④ consider the dataset 10, 12, 13, 14, 15, 16, 17, 18, 20, 250. Identify outlier using the IQR method

sol/ data : 10, 12, 13, 14, 15, 16, 17, 18, 20, 250

Find Q_1

lower half

10, 12, 13, 14, 15

$$\therefore Q_1 = 13$$

Find Q_3

upper half

16, 17, 18, 20, 250

$$\therefore Q_3 = 18$$

IQR

$$IQR = Q_3 - Q_1$$

$$= 18 - 13$$

$$IQR = 5$$

calculate lower boundary

$$L = Q_1 - 1.5(IQR)$$

$$= 13 - 1.5(8)$$

$$= 13 - 12$$

$$L = 1$$

calculate upper boundary

$$U = Q_3 + 1.5(IQR)$$

$$= 18 + 1.5(8)$$

$$= 18 + 12$$

$$U = 30$$

Identify the outlier

n value : 25.5 < 25.5

Here

$$50 > 25.5$$

∴ 50 is an outlier

outlier treatment

$$\text{median} = \frac{15+16}{2} = 15.5$$

50 can be replaced by 15.5

$$\therefore \text{outlier} = 50$$

Q. Sample the exam scores are 40, 42, 45, 46, 48, 50, 52, 90

Identify outlier using z-score method

sol. given

40, 42, 45, 46, 48, 50, 52, 90

given

$$\mu = 51.625$$

$$\sigma = 15.09$$

$$\text{Formula: } z = \frac{x - \mu}{\sigma}$$

$$\text{for } x = 90; z = \frac{90 - 51.625}{15.09}$$

$$z = \frac{38.375}{15.09}$$

$$z \approx 2.45$$

Apply result

$$1 < 2.45$$

$$2.45 < 3$$

∴ 90 is not an outlier according to the ± 3 rule

6) consider 2, 2, 3, 4, 4, 5, 5, 6, 10

skewness

given

2, 2, 3, 4, 4, 5, 5, 6, 10

calculate mean

$$\text{mean} = \frac{\sum x}{n}$$

$$\sum x = 2 + 2 + 3 + 4 + 4 + 5 + 5 + 6 + 10 = 41$$

$$n = 9$$

$$\text{mean} = 44/9$$

$$\text{mean} = 4.67$$

median

$$\frac{n+1}{2} = 5$$

5th value

$$\text{median} = 4$$

compare mean and median

$$\text{mean} = 4.67$$

$$\text{median} = 4$$

$$\text{mean} > \text{median}$$

the distribution is positively skewed

$\therefore$ distribution is positively right skewed

7) The waiting time for a bus is uniformly distributed b/w 0 and 20 minutes. Find the probability that a passenger waits between 5 and 12 minutes

sol 1 to find:

$$P(5 < x < 12)$$

Formula

$$P(a < x < b) = \frac{b-a}{b-a}$$

$$\therefore a = 0, b = 20, c = 5, d = 12$$

S. betw

$$P(5 < x < 12) = \frac{12-5}{20-0}$$

$$= \frac{7}{20}$$

$$= 0.35$$

$$= 0.35$$

convert to percentage

$$0.25 \times 100 = 25\%$$

$$P(S < 1.5) = 0.25$$

25%

- ⑧ The heights of students are normally distributed with $\mu = 170 \text{ cm}$, $\sigma = 10 \text{ cm}$. Find the z-score for a student whose height is 185 cm.

Heights are normally distributed with

$$\mu = 170 \text{ cm}$$

$$\sigma = 10 \text{ cm}$$

Find the z-score $x = 185 \text{ cm}$

Formula

$$z = \frac{x - \mu}{\sigma}$$

Substitute

$$z = \frac{185 - 170}{10}$$

$$z = 15/10$$

$$z = 1.5$$

Final Answer

$$\boxed{z = 1.5}$$

The student height is 1.5 standard deviation above

the mean

- ⑨ using the previous example Find the probability that a randomly selected student has a height less than 165 cm.

From table:

$$z = 1.5 \Rightarrow 0.9332$$

use the standard d. normal

$$\text{for } z = 1.5$$

$$P(Z < 1.5) = 0.9332$$

convert to percentage

$$0.9332 \times 100 = 93.32\%$$

Serial answer:

$$DC \times 21851 = 0.9332$$

$$93.32\%$$

Interrelational

Approximately 93.32% of student are expected

so now

rightly

below 105 cm

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